ZOOMACHIA

ANIMAL BATTLES BASED ON SCIENCE FACTS

EP01 · Siberian Tiger vs Ussuri Brown Bear

Verdict report · every number, source and score

This report contains the result of the episode.

Science Standard v2.2 · AI-simulated duel built from published data · zoomachia

How we score at Zoomachia

1. Every number has a source and an Evidence Score

Each fact in an episode (a mass, a bite force, a speed, a claw length, a field record) comes from a published source. We rate how solid it is with an **Evidence Score from 0 to 100**, then a plain-words band:

Band	Score	What it usually means
Solid	80–100	measured on wild animals, peer-reviewed, a good sample
Good	60–79	a reliable agency or reference work, or a smaller measured sample
Fair	40–59	an estimate (e.g. from skull measurements) or a single report
Weak	0–39	a descriptive account; we say so, and it weighs less

The score depends on the type of source (study, agency, reference work, descriptive page), how the number was obtained (measured, estimated, reported), the sample, and whether other sources agree.

2. Fighter cards come from checklists, the same for every animal

Each animal gets ratings from 1 to 99 for **size, power, speed, weapons, defense, stamina, fight IQ and instinct**, plus **terrain** for the battlefield. Every rating is built from a sourced checklist (for example: body mass for size, bite and strike force for power). The checklists are the same for every animal.

As used in episode 1 (Science Standard v2.2), two parts of the cards also use the records of these two species meeting each other:

- **Instinct** includes the share of encounters each species started: 13 of 21 for the tiger, 8 of 21 for the bear (field records from the Sikhote-Alin).
- **Each secret weapon** scores +25 because each species has killed the other on record.

The terrain also changes with the battlefield. **From episode 2 the model changes:** a card is computed before any duel and never depends on the opponent. See Appendix B, "We improved the model after episode 1". Episode 1 keeps the model it was made with.

3. The verdict comes from 20,000 simulated fights

We do not pick the winner. A fixed program plays **20,000 fights** between the two cards, exchange by exchange: who acts first, which strengths meet, fatigue, the battlefield and each fighter's **secret weapon** (a documented move that can turn a fight). The winner is the animal that wins most of them. The random seed is public, so the same inputs always give the same result.

- **Close Call**: when the split is between 48 % and 52 %, the result is too close to call on the main run, and a **Decider** (a second, independent set of runs) settles it.
- **The fight you watch** is one of the simulated fights where the overall winner wins, chosen so it shows the most likely course of events (for example, the animal that usually strikes first strikes first).
- **The Wildcard** changes one documented condition (a season, a terrain) and the whole simulation is run again.

4. What this is not

It is not a prediction of what any individual animal would do, and it is not real footage. Real fights between two prime adults are rarely or never recorded; the simulation is built from what each animal can really do. Found a better source? Tell us in the comments: accepted corrections are credited in the changelog of the report.

VERDICT DOSSIER: EP01, Siberian Tiger vs Ussuri Brown Bear

Zoomachia · Science Standard v2.2 (engine SHA-256 8ab44c8a…; v1.0 locked 17/10/2026, v2.0 and v2.1 18/10/2026, v2.2 19/10/2026; the main verdict is identical under all four, and the Wildcard result is v2.2, see §6) · battlefield: late-autumn taiga, first snow (Sikhote-Alin, Russian Far East) · fighters: typical prime wild adult males · 18/10/2026.

This page lists the facts behind the verdict, how each one was scored, and how the verdict was computed. No story, no dramatisation. How the scores and the verdict work: *How we score* (at the start of this report). Every number can be checked against the sources at the bottom.

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1. Why this match-up is real

*The Siberian tiger in this report is the **Siberian (Amur) tiger, Panthera tigris altaica**: the same animal. Some sources below use the name Amur.* The two species share the same forests and do meet. A 1992–2013 study in Sikhote-Alin Reserve found 763 tiger kills; bears made up 2.2 % of them, and bear remains appeared in 24.3 % of tiger scats in autumn [S1]. Brown bears also took tiger kills by force 11 times [S1]. In 45 recorded tiger–brown bear encounters reviewed in the same paper, tigers started 13 and bears 8; the bear died in 51 %, the tiger in 27 %, and both left in 22 %; brown bears have killed 12 tigers, 6 of them adults [S1]. **What this record cannot tell us:** it mixes sexes and ages. Of the 9 brown bears found killed by tigers, 6 were adult females, 1 was a young male and 1 a cub; the only adult male was scavenged, not killed [S1]. A prime male against a prime male is the rare case, and the one we model.

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2. The facts (with Evidence Scores)

Metric	Siberian tiger	Evidence	Ussuri brown bear	Evidence
Body mass	**180 kg** (165–195) [S5, S6]	80 · Solid	**267 kg** (217–317, n = 24) [S2]	92 · Solid
Canine bite force	**≈ 1,487 N** (1,264–3,375), scaled from 1,525 N at 187 kg (estimated mass) [S3, S4]	40 · Fair	**≈ 1,225 N** (1,041–1,627), scaled from 751 N at 129 kg (estimated mass) [S3, S4]; for reference, not counted: 1,410 N for a grizzly by the same skull model [S17, quoted in S18]	40 · Fair
Top burst speed	**56 km/h** (48–65) [S8, S9]	43 · Fair	**48 km/h** (40–64) [S8, S11]	65 · Good
Front paw pad width	**11.4 cm** (n = 12) [S7]	85 · Solid	**17.4 cm** (n = 28) [S2]	92 · Solid

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3. The Fighter Cards

	Siberian tiger	Evidence	Ussuri brown bear	Evidence	How it was scored
OVR	**61**	59 · Fair	**61**	72 · Good	weighted card average (Standard §3.5); evidence weighted the same way
SIZE	62	80 · Solid	66	92 · Solid	formula from mass
POWER	69	40 · Fair	66	40 · Fair	formula from bite force (both scaled to the fighter's mass by mass^0.67, Standard §2)
SPEED	52	43 · Fair	46	65 · Good	formula from top speed
DEFENSE	50	56 · Fair	55	85 · Solid	tiger: winter coat 40 (56) + fat layer 10 (56) [S15] · bear: winter coat 40 (56) [S16] + fat 10 (100) + shoulder hump 5 (100) [S16, S11]
WEAPONS	75	65 · Good	65	85 · Solid	tiger: long canines 30 (73) + retractable hooked claws 20 (73) [S10, S14] + forelimb grapple 10 (56) [S10] + nape/neck killing bite 15 (56) [S10] · bear: long canines 30 (56) [S12] + front claws 5.9–6.5 cm on average in adult males, max 8.5 cm, 15 (100) [S2, S12, S16] + one-blow strike 20 (100) [S13, S19]
STAMINA	35	56 · Fair	70	56 · Fair	tiger: ambush hunter 30 (56) [S10] + travels 16–32 km a night 5 (56) [S14] · bear: endurance fighter 70 (56) [S13]
FIGHT IQ	60	79 · Good	50	82 · Solid	tiger: picks vulnerable bears (82) [S1], stalks using cover (73) [S10, S14], waited a day at a den (82) [S1] · bear: displaces tigers from kills (82) [S1], caches food (82) [S1]
INSTINCT	75	82 · Solid	55	82 · Solid	share of encounters started (13/21 vs 8/21) (82) [S1] + territory (82) [S1] + confronts large predators (82) [S1]
TERRAIN: COLD	80	80 · Solid	80	68 · Good	both: insulating coat (56) [S15, S16] · tiger active all winter 20 (100) [S1, S6], foot loading 1.39 → +10 (83) · bear dens in mid-winter 10 (56) [S13], foot loading 0.88 → +20 (92)
Secret Weapon	Nape Strike 75		Hammer Paw 75		both have killed the other species on record (+25) [S1] and can end a fight alone (+10) [S10, S13, S19]
INITIATIVE	**70.1**	70 · Good	51.3	77 · Good	the tiger is likely to strike first

Evidence of each trait in brackets after its points. Evidence Scores are shown for transparency only: they never enter the simulation (Standard §1.1).

Foot loading = mass / pad width^2: the bear is heavier but has much wider paws, so it presses less on the snow than the tiger (0.88 vs 1.39).

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4. The verdict (20,000 simulated fights)

The fight came out 50/50. Over 20,000 simulated fights, each animal won half of them: too close to call. When the split falls between 48 % and 52 % (a Close Call), a second set of runs, the **Decider**, settles it. The Decider gives a little more weight to each fighter's signature strengths, the same way for every duel:

	Main run	Decider	Formula
Secret Weapons (rating)	Nape Strike 75 · Hammer Paw 75	same ratings	chance to land when its owner wins an exchange at 60 health or less = rating / 200
Secret Weapon damage	+22	**+33**	+22 × 1.5
Battlefield (TERRAIN) rating	tiger 80 · bear 80	same ratings	
Weight of TERRAIN in the fight	10 % of the exchanges	**20 %**	10 % × 2
Result	**50/50**	**Bear 54 % · Tiger 46 % → BEAR WINS**	54 % is outside the 48–52 % zone: a clear gap

What happened in the fights	Value
Tiger's Nape Strike	lands in 36.9 % of fights; tiger then wins 85.4 % (29.7 % when it does not land)
Bear's Hammer Paw	lands in 56.3 % of fights; bear then wins 78.4 % (12.9 % when it does not land)
Fights won by K.O.	99.6 %
Typical length	7 exchanges
Tiger strikes first	85.4 %
Favourite was behind at some point	51.3 %

In the video the result is told in words only ("as close as a fight can get", then the bear's strengths): the video shows no model percentage (Standard §5). This page gives the numbers behind it.

What decides it: the tiger's edges are WEAPONS, INITIATIVE and FIGHT IQ: it usually lands first and must finish early. The bear's edges are SIZE, DEFENSE (it soaks more of each hit) and STAMINA: every exchange the fight lasts tips it the bear's way. They cancel out almost exactly, so the fight is decided by whichever **Secret Weapon lands**. The bear's Hammer Paw lands more often (it usually falls behind early, which puts it in range, and it is still fresh late in the fight), which is why the Decider gives the bear the win.

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5. What would change the verdict (sensitivity checks)

Change	Tiger wins
Baseline (another seed)	49.9 %
Tiger at the low / high end of its mass range (165 / 195 kg)	47.0 % / 50.5 %
Bear 1 SD lighter (217 kg)	55.0 %
Bear 1 SD heavier (317 kg)	48.1 %
Tiger 200 kg	49.2 % (heavier tiger sinks more in snow: foot loading crosses 1.5)
Tiger bite from the lever-model method (3,375 N) [S4]	55.6 %
Both bites from the lever-model method [S4]	54.4 %
No encounter data (both start 50 % of fights)	44.2 %
If the bear could also pin large prey down with its front legs (hold it under one forepaw while it bites or strikes, which keeps the prey from twisting away)	31.6 %. **Not applied: no source found yet** that wild brown bears do this to large prey
Bear without its long claws (if they were under 5 cm)	75.4 %. **Not applied: claws measured at 5.9–6.5 cm [S2]**
Tiger without retractable hooked claws / without its neck-bite technique	22.0 % / 23.8 %
Bear without its one-blow strike	84.5 %

Method: one change at a time, same random seed as the main run. Mass changes shift the whole range; lever-model bites use the top of each range as the central value; "another seed" adds `|alt` to the seed. Every ticked trait was also removed one at a time (full table on request).

Tipping points: two of the bear's traits decide this duel. Without either one, the tiger becomes the clear favourite:

- **Long claws ≥ 5 cm**: without them, tiger **75 %** (CLEAR). Evidence **100 · Solid**: measured on adult male Ussuri brown bears, front claws 5.9–6.5 cm on average along the outer curve (n = 12), up to 8.5 cm (Seryodkin 2025, Table 2) [S2], confirmed by [S12, S16].
- **One-blow power strike**: without it, tiger **84.5 %** (DOMINANT). Evidence **100 · Solid**: a brown bear can kill a cow with one blow [S13]; it can kill prey with a strike of its forepaw, often enough to break the spine [S19, Finnish state wildlife page]. Caveat: [S19] is a descriptive page, not a study, and no field study measuring or counting such blows was found; a grizzly ecology website (Mostly Natural Grizzlies, not scored) describes killing paw blows as rare, bites being the usual method. This remains the point a better source could still move.

Honest reading: with today's evidence this fight is a coin flip: even another random seed puts the bear marginally ahead before the Decider. Single traits move it a lot (one item is worth 15 to 35 points of win probability), which is why each one is sourced. The biggest open question is still whether wild brown bears are documented pinning large prey down with their front legs; that one item would make the bear the clear favourite.

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6. Wildcard: SEASON SHIFT

The card: one external variable changes, the **season**. Same two animals, same look, same fighter files; the fight happens in **early spring, just after the bear leaves its den**, instead of late autumn. Only the sourced, quantified consequences of that season are applied, to both animals, whatever their effect on the result (Standard v2.2 §5).

Consequence	Value	Source	Evidence	Card change
Bear's body mass after hibernation	measured: northern Scandinavian males weigh 201 kg in spring vs 273 kg in autumn (73.6 %; 1,771 bears in the study). The loss is probably smaller here (shorter denning, ≈ 5 months; 267 kg is not a late-autumn mass), so the uncertainty goes into the range: **197 kg** (the full loss) **to 232 kg** (half of it), **central 214.5 kg** (midpoint; was 267 kg, 217–317)	Swenson et al. 2007, *Oecologia* 153: 37–47	92 · Solid (the measured ratio)	bear SIZE 66 → **64**
Bear's fat reserve spent over winter	brown bears lost 32 ± 10 % of fall mass over 208 days, 56 % of it lipid (55 ± 22 kg); fat supplied 88 % of winter energy	Hilderbrand et al. 2000, *J. Wildl. Manage.* (doi 10.2307/3802988); fat built before winter [S16, S11]	100 · Solid	DEFENSE item "thick fat layer" removed: bear DEFENSE 55 → **45**
Snow almost melted at den exit (v2.2)	bears usually do not leave the den until the snow has almost melted, so there is no deep snow to sink into: the paw-width (foot-loading) part of the snow rating no longer counts, **for both animals**; their other snow items stay	Seryodkin, *Okhrana Dikoy Prirody* 24; dissertation	82 · Solid	tiger TERRAIN 80 → **70**, bear 80 → **60**

Caveats: the fat figures are from adult females in Alaska (no male or Russian figure found). With the full mass loss as the central value the tiger wins 73.7 %; with only half of it, 71.3 %: the result does not hang on the exact loss. Two possible spring effects were looked for and not applied, because no source gives them a number: a spring snow crust in the Sikhote-Alin, and the tiger's condition at the end of winter.

Result (engine v2.2, seed `amur_tiger|ussuri_brown_bear|COLD|SEASON_SHIFT`, 20,000 fights): **Tiger 80.1 % · Bear 19.9 %, DOMINANT. The verdict FLIPS: the tiger wins.** (Without the snow consequence, v2.1: 72.6 %, CLEAR.) Bear OVR 61 → 59. Tiger strikes first in 86.4 % of fights (85.4 % in the main fight: the card changes none of the first-strike inputs, which are aggression, speed, fight IQ and ambush). **Story fight chosen among the tiger's wins where it strikes first (its most likely opening)**, with the same other rules as before (the tiger's Nape Strike lands after at least one exchange, the bear's Hammer Paw lands, most dramatic; 1,726 fights qualified): the tiger opens (X1), the Hammer Paw lands (X5: tiger 66 → 33), then the Nape Strike drops the bear 42 → 10 (X7) before the K.O. The main fight shown in the video follows the same rule: the tiger strikes first there too.

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7. Limits

Real fights between prime males are almost never witnessed; this is a model of documented abilities. Bite forces are skull-model estimates (Fair evidence); no published measurement exists for the force of a bear's paw blow. Rules and limits: *How we score*.

Calibration parameter, not a measurement: the Secret Weapon's +22 damage. When a Secret Weapon lands it adds 22 damage to the hit (a normal hit is 10–28 before absorption, so a Secret Weapon hit is 32–50: a third to half of a fighter's health in one exchange). This value was fixed when the Standard was first written (v1.0) and tested on the calibration bench; it is a setting of the model, not a measured quantity. How much it matters here (same seed, 20,000 fights):

Secret Weapon damage	Tiger (main run)	Decider	Winner
+15	51.3 %	bear 52.9 %	bear
+22 (used)	**50.2 %**	**bear 54.4 %**	**bear**
+30	48.5 %	bear 57.3 %	bear
+44	44.7 % (narrow, no Decider needed)	not run	bear

The bear wins in every case; only how close it is moves.

In the video, the K.O. exchange is shown as a near-K.O. hit (Standard v2.2 §5): in X8, the bear drives the spent tiger into the drift (tiger 18 → 8, bear 17 → 13), then the stylised finisher takes it to 0. Staging only: no number changes.

Appendices at the end of this report: A. The ratings example (cougar, fisher, porcupine) · B. We improved the model after episode 1 · C. How the Wildcard fight and the health bars are shown.

Sources

- [S1] Seryodkin, Miquelle, Goodrich, Kostyria & Petrunenko (2018). Interspecific relationships between the Amur tiger and brown and Asiatic black bears, *Biology Bulletin* 45. https://link.springer.com/article/10.1134/S1062359018080149
- [S2] Seryodkin (2025). Morphometric characteristics of brown bears captured in the Middle Sikhote-Alin, *Pacific Geography* 2025(1): 63–72. https://www.tigdvo.ru/assets/files/publications/Pacific_Geography_journal/2025_1/63-72.pdf
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- [S5] WCS / NFWF. Ecology of Amur Tigers in Primorye, final report. https://www.nfwf.org/sites/default/files/finalreports1/16403_tigerleopardFINAL.pdf
- [S6] Center for the Study and Conservation of the Amur Tiger. Who is the Amur tiger? https://amur-tiger.ru/en/about-tiger

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- [S12] International Association for Bear Research and Management. Brown bear. https://www.bearbiology.org/the-eight-bear-species/ursus-arctos-brown-bear/
- [S13] Animal Diversity Web. Ursus arctos. https://animaldiversity.org/accounts/Ursus_arctos/
- [S14] Animal Diversity Web. Panthera tigris. https://animaldiversity.org/accounts/Panthera_tigris/
- [S15] Zoo Berlin. Amur tiger. https://www.zoo-berlin.de/en/explore-the-zoo/amur-tiger
- [S16] Brookfield Zoo Chicago. Brown bear. https://www.brookfieldzoo.org/animals/brown-bear
- [S17] Christiansen & Wroe (2007). Bite forces and evolutionary adaptations to feeding ecology in carnivores, *Ecology* 88(2): 347–358. https://doi.org/10.1890/0012-9658(2007)88[347:BFAEAT]2.0.CO;2 (reference only: same skull model as [S3], specimen mass not available, not counted)
- [S18] White, Gunther & van Manen (eds.) (2017). Yellowstone Grizzly Bears: Ecology and Conservation of an Icon of Wildness, Yellowstone Forever / National Park Service. https://www.nps.gov/yell/learn/nature/upload/Yellowstone_Grizzlies_Web.pdf
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Appendices

Appendix A. The ratings example: cougar, fisher and porcupine

The video's ratings segment uses this example to show that a card says who should win, not who will. ***It uses an example-specific design*** *(a one-off choice for this example): it is not part of the Science Standard, and it is not used to score any duel.*

The triangle

All three are North American species. Each claim is about the same species as its source (no mixing of species across sources).

- **A cougar can kill a fisher.** Of 101 fisher carcasses studied in California, 62 had been killed by other predators. Mountain lions killed 20 of them, identified by predator DNA in the saliva left on the carcasses [A1].
- **A fisher hunts porcupines.** See below.
- **The porcupine can beat a cougar.** Of 59 cougar deaths documented in the Southern Yellowstone Ecosystem (2001–2015), 2 followed hunts of North American porcupines (*Erethizon dorsatum*); that is 9 % of the deaths of young cougars [A2]. Earlier, 3 cougar kittens were killed by quills [A3].
- These records concern young cougars, so the video says "some cougars".
- A record of lions killed by quills concerns another species (the African crested porcupine), so it is not used.
- The cougar card on screen is the calibration-bench card, Standard v2.2: overall rating 52.

What the field records say about the fisher and the porcupine

- **How the fisher attacks.** The fisher (*Pekania pennanti*, a forest member of the weasel family) kills porcupines with repeated bites to the face, which has few quills. It attacks from the front, avoids the tail, and works until it can flip the porcupine over [A4, A5].
- **How long it takes.** An attack can last about half an hour [A4].
- **The effect on whole forests.** Fishers were reintroduced to Wisconsin forests to reduce porcupine damage, and porcupine numbers fell where fishers spread [A6]. Fishers "can effectively control porcupine populations" [A5].
- **What is not known.** No published source gives an attack success rate, so the video never says the fisher "usually wins".

The two example cards

	Fisher (adult male)	Porcupine (adult male)
OVR	**38**	**40**
POWER	38 (bite ≈ 113 N, general model)	44 (≈ 183 N, general model)
SIZE	29 (4.8 kg, measured)	36 (7.9–11.8 kg, measured)
SPEED	38 (general model)	13 (judgement: "slow, lumbering")
DEFENSE	40 (winter fur)	70 (quill coat, example-specific judgement)
WEAPONS	40 (canines + climbing claws 5 + killing bite)	35 (incisors + quill tail strike)
STAMINA / FIGHT IQ / INSTINCT	40 / 40 / 50	30 / 30 / 50

Honest limits

- **Two judgements put the fisher's card below the porcupine's**, and both are marked example-specific: the quill coat at 70, and the fisher's claws at 5.
- **The fisher's claws are scored conservatively.** The sources describe sharp, retractable climbing claws used to hold prey. Under the fighter checklist they would score 20 instead of 5, which would give the fisher 41 against the porcupine's 40. We kept 5 because the fisher kills with its bite, not with its claws, so the cards on screen stay 38 / 40.
- **This is a predator and its prey, not a duel.** The video's wording rests on the field records above, never on a simulated percentage.

Sources for Appendix A

- [A1] Wengert et al. (2014), *Journal of Wildlife Management* 78: 603–611: 101 fisher carcasses in California, predators identified by DNA.
- [A2] Elbroch, Hoogesteijn & Quigley (2016), *Canadian Field-Naturalist* 130(1): 53–55. https://www.canadianfieldnaturalist.ca/index.php/cfn/article/view/1793
- [A3] Robinette et al. (1959), as cited in [A2].
- [A4] Washington Department of Fish and Wildlife (1998), *Washington State Status Report for the Fisher*, citing Powell (1993), *The Fisher: Life History, Ecology, and Behavior*. https://wdfw.wa.gov/sites/default/files/publications/00227/wdfw00227.pdf
- [A5] Animal Diversity Web, University of Michigan: fisher (*Pekania pennanti*) and North American porcupine (*Erethizon dorsatum*) accounts.
- [A6] Wisconsin Department of Natural Resources, fisher history (releases 1956–1967). https://asset.library.wisc.edu/1711.dl/WMPS5J2O6OLDZ87/E/file-c71a6.pdf

Appendix B. We improved the model after episode 1

After episode 1, we improved the way we score animals. From episode 2:

- **A fighter's card no longer depends on its opponent.** In episode 1, part of the tiger's and the bear's ratings came from records of the two species meeting (see "How we score", §2). From now on, every card is computed before a duel is chosen.
- **A returning animal keeps its card.** The same animal brings the same card to every duel.
- **Only the battlefield changes.** The terrain rating is the one part of a card that depends on where the duel takes place.

Other improvements are announced in the report of the episode that first uses them. Episode 1 keeps the model it was made with (Science Standard v2.2).

Appendix C. How the Wildcard fight is shown

Video models cannot follow eight fast exchanges from 100 to 0 coherently. So the Wildcard shows its decisive beats as clips:

- the tiger's opening strike;
- the Hammer Paw;
- the Nape Strike;
- the bear folding.

The other exchanges (X2–X4, X6) appear as health-bar changes over impact frames. Every number on the bars is the engine's story fight, unchanged. That story fight was chosen among the tiger's wins in which it strikes first, its most likely opening.

Health bars in both fights. Each exchange shows the winner's hit first, with its full damage on the visible contact. The loser's counter (the smaller damage it still lands) then appears as its own hit, on the loser's next visible move, or as a short drain right after the hit when the picture shows no counter. The totals are exactly the engine's.